Multi-Cancer Image Classification

Neural Networks and Genetic Algorithms - three-week progress update

CURRENT STAGE **Data audit and leakage-aware split design before final model benchmarking**

130,000	8	26	5
verified images	top-level groups	leaf classes	imaging modalities

What has been completed

Dataset and label audit

- All 130,000 accepted images were decoded; no corrupt image was found.
- The dataset contains 8 parent folders, 26 leaf labels and 5 acquisition modalities.
- File format, size, channel and label-taxonomy checks were completed.
- 1,318 exact-duplicate groups were identified, containing 2,673 files.

Leakage-control pipeline

- A 1,342-pair calibration set covers all 26 leaf classes.
- 671 development pairs were scored; 671 holdout pairs remain sealed.
- Implemented pHash, ORB/RANSAC, phase correlation, ECC, SSIM and NMI evidence.
- 23 automated tests pass in the Python 3.10 environment.

Why final training has not been reported yet

The downloaded dataset was already augmented. A naive random image-level split can place transformed versions of the same source image in both training and test sets, producing unrealistically high accuracy. Therefore, the final model comparison is intentionally postponed until the duplicate-family method and leakage-aware split are validated.

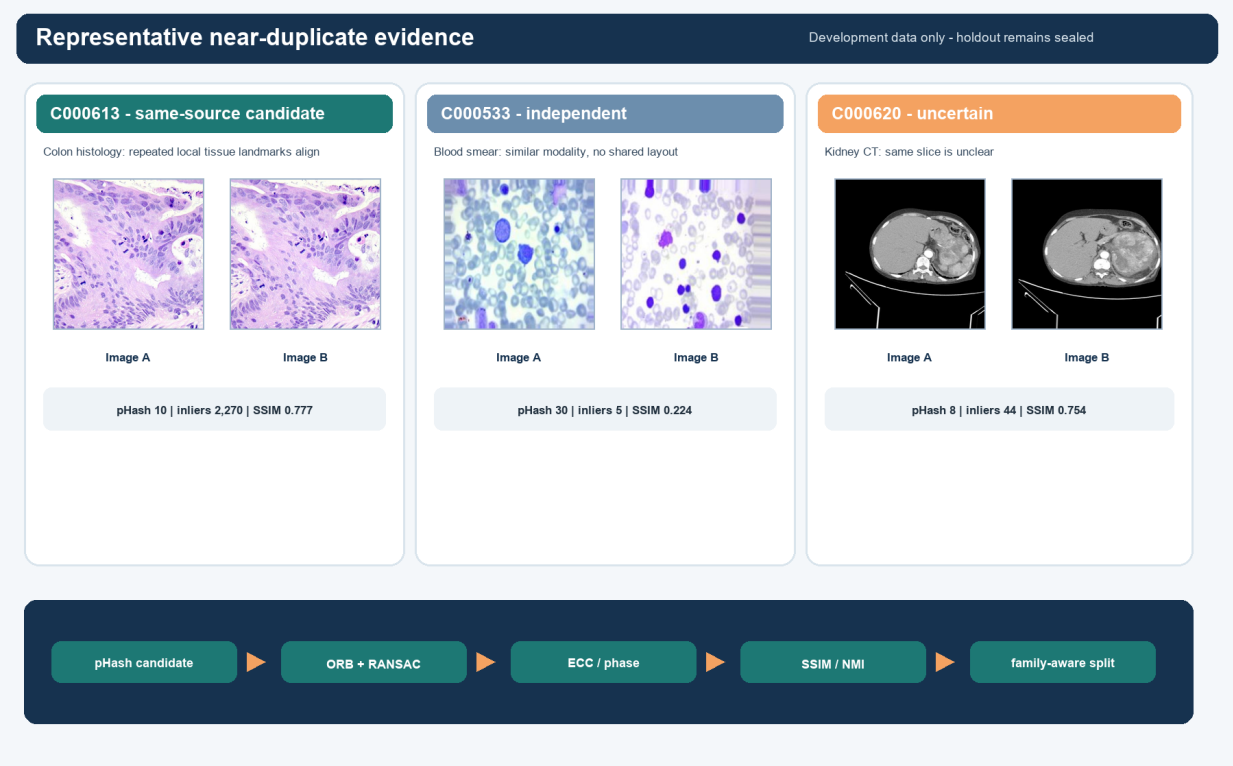
Project path

1. Audit
2. Calibrate
3. Freeze rule
4. Holdout test
5. Model benchmark

Scope note: the required 8 top-level group classification will be reported. The 26 leaf labels are being studied as a structured extension and can be mapped back to the 8 parent groups. No clinical diagnosis or cancer-stage claim is made.

Representative evidence and current findings

Examples below show why one similarity score is not sufficient and why an explicit uncertain region is required.



Development findings

- 15 visually accepted natural candidates: 449-2,360 ORB/RANSAC inliers.
- 123 independent pairs: 3-17 ORB/RANSAC inliers.
- C000620 falls between both groups and remains uncertain.
- 270/520 controlled positives had pHash distance above 12.

Next controlled steps

- Freeze a conservative multi-evidence duplicate rule using development only.
- Evaluate the frozen rule once on the sealed holdout set.
- Build duplicate families and a leakage-aware 70/15/15 manifest.
- Compare SmallCNN, ResNet18, DenseNet121 and EfficientNet-B0 on the same split.

Interpretation limit: development separation is encouraging but is not holdout performance and is not classifier accuracy. The final claim will be made only after the frozen method is evaluated on unseen holdout pairs and the model is tested on a leakage-aware split.

Code and documentation: GitHub repository. Dataset source: Obuli Sai Naren, Multi Cancer Dataset (Kaggle).

How the duplicate evidence works

No single number decides whether two images are copies. The pipeline combines complementary evidence so that a similar-looking image is not automatically treated as the same source.

Method	What it measures	Why it is used	How to read it
pHash	A compact fingerprint of the overall visual layout.	Finds candidate pairs quickly.	Small distance means globally similar; it is not proof.
ORB	Distinctive local points such as edges, nuclei or tissue marks.	Checks whether local details can be matched.	More reliable matches are stronger evidence.
RANSAC	Whether ORB matches agree with one rotation/shift/scale transform.	Rejects accidental matches.	An inlier supports one coherent geometry.
ECC / phase	Global alignment and displacement between two images.	Supports low-texture MRI/CT images.	A high score means alignable intensity structure, not identical origin.
SSIM	Local structural similarity after alignment.	Checks shape and contrast together.	Closer to 1 means more structurally similar.
Correlation	Whether intensity changes rise and fall together.	Adds a global intensity-pattern check.	Closer to 1 means stronger shared pattern; alone it is insufficient.
NMI	Shared information between image intensities.	Helps when brightness or contrast differs.	Supportive evidence, interpreted with the other measures.

Reading the current numbers

- Same-source reference examples: ORB/RANSAC inliers 449-2,360, SSIM 0.777-0.969 and correlation 0.959-0.999.
- Independent-looking examples: inliers 3-17, SSIM 0.012-0.742 and correlation -0.299-0.837.
- C000620 is between these ranges, so it remains uncertain rather than being forced into either class.
- These ranges describe development evidence only; the frozen rule must still be evaluated once on sealed holdout data.

Important scope note

This audit protects the future train/validation/test split from near-duplicate leakage. It is not a cancer diagnosis, not a patient-identity test, and not a model-accuracy result. The holdout remains sealed until the rule and its configuration are frozen.